
Chapter 9: Numerical Methods

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Not every equation can be solved exactly using algebra — $x^3 - 4x - 2 = 0$, for instance, has no “nice” solution you can write down using the techniques from earlier chapters. Numerical methods are systematic ways of finding increasingly accurate **approximate** solutions to such equations. The methods here are genuinely practical — they’re essentially what a calculator or computer does internally to solve equations — and they connect directly back to differentiation (Chapter 7), since the Newton–Raphson method uses the gradient of a curve to home in on a root.

9.1 Locating roots by a change of sign

Concept

If a continuous function f is negative at one point and positive at another, it must cross zero somewhere in between (this is really just the intuitive idea that you can't get from below the x -axis to above it without crossing it, for a curve with no breaks or jumps). This gives a simple, robust way to confirm a root exists in an interval, and to narrow that interval down.

Procedure

1. Evaluate $f(x)$ at two values, a and b , where $a < b$.
2. If $f(a)$ and $f(b)$ have opposite signs (one positive, one negative), then by the change-of-sign principle, a root lies somewhere in (a, b) — as long as f is continuous on that interval, which is true for every function in this course.
3. To narrow down the root's location, evaluate f at points within (a, b) (commonly the midpoint) and repeat: whichever half now shows a sign change contains the root.

Worked example

EXAMPLE 9.1

Show that the equation $x^3 - 4x - 2 = 0$ has a root between $x = 2$ and $x = 3$, then narrow the root down to one decimal place.

Step 1 Let $f(x) = x^3 - 4x - 2$. $f(2) = 8 - 8 - 2 = -2$ (negative). $f(3) = 27 - 12 - 2 = 13$ (positive) (M1).

Step 2 Since $f(2) < 0$ and $f(3) > 0$, and f is continuous (it's a polynomial), there is a root between $x=2$ and $x=3$, by the change of sign (A1).

Step 3 Narrowing down: try $f(2.2) = 10.648 - 8.8 - 2 = -0.152$ (negative); $f(2.3) = 12.167 - 9.2 - 2 = 0.967$ (positive) (M1). So the root lies between 2.2 and 2.3.

Step 4 Since $f(2.2)$ is negative and $f(2.3)$ is positive, check the midpoint 2.25 to decide which side of it the root falls on before rounding to 1 d.p.: $f(2.25) = 2.25^3 - 4(2.25) - 2 = 11.390625 - 9 - 2 = 0.390625$ (positive).

Answer Since $f(2.2) < 0$ and $f(2.25) > 0$, the root lies in $(2.2, 2.25)$, so to 1 d.p. the root is $x = 2.2$ (M1, A1 — the midpoint check is required for a fully rigorous conclusion; comparing which function value happens to be closer to zero is not valid justification on its own).

COMMON MISTAKE

Concluding “there is a root” from a sign change without checking that the function is continuous on the interval — for functions with asymptotes (like $\tan x$, or $1/x$), a sign change can occur *without* a root, simply because the function jumps from $+\infty$ to $-\infty$ across a discontinuity. Always check (or state) continuity before applying this method.

9.2 Fixed-point iteration

Concept

Fixed-point iteration solves $f(x) = 0$ by first rearranging it into the form $x = g(x)$, then generating a sequence of approximations using the rule $x_{n+1} = g(x_n)$, starting from some initial guess. If the sequence converges (settles down to a single value), that limiting value L satisfies $L = g(L)$, which means $f(L) = 0$ — exactly the root we wanted. This is a recurrence relation (4.4) applied to root-finding.

Procedure

1. Rearrange $f(x) = 0$ into the form $x = g(x)$ (there are often several ways to do this — not all of them will converge, see 9.3).
2. Choose a starting value x_0 (often based on a change-of-sign search from 9.1).
3. Iterate: compute $x_1 = g(x_0)$, then $x_2 = g(x_1)$, and so on, continuing until successive values agree to the required number of decimal places.
4. State the root to the required accuracy, and ideally verify by substituting back into the original equation.

Worked example

EXAMPLE 9.2

Show that $x^3 - 4x - 2 = 0$ can be rearranged into the iteration formula $x = (4x+2)^{1/3}$, and use it with $x_0 = 2$ to find the root to 3 decimal places.

Step 1 Rearranging: $x^3 = 4x + 2 \implies x = (4x+2)^{1/3}$ (M1, algebraic rearrangement).

Step 2 Iterating from $x_0 = 2$: $x_1 = (4(2)+2)^{1/3} = 10^{1/3} = 2.1544$. $x_2 = (4(2.1544)+2)^{1/3} = 2.1979$. $x_3 = 2.2098$. $x_4 = 2.2131$. $x_5 = 2.2140$. $x_6 = 2.2142$ (M1, A1 for the sequence of values, calculator-generated).

Answer Since x_5 and x_6 agree to 3 decimal places (2.214), the root is $x \approx \mathbf{2.214}$ (A1). (Compare with 9.1's bracket of $[2.2, 2.3]$ — consistent.)

n	0	1	2	3	4	5	6
x_n	2	2.1544	2.1979	2.2098	2.2131	2.2140	2.2142

COMMON MISTAKE

Stopping the iteration too early, before successive terms have actually converged to the required accuracy — always continue until at least two consecutive iterates agree to one more decimal place than what's being asked for, as a safety margin.

9.3 Cobweb and staircase diagrams, and convergence

Concept

A cobweb or staircase diagram visualises the fixed-point iteration process geometrically, by plotting both $y = g(x)$ and $y = x$ on the same axes and tracing the sequence of iterates between them. Whether the resulting picture looks like a **staircase** (converging steadily from one side) or a **cobweb** (spiralling in from both sides alternately) — or whether it diverges entirely — depends on the gradient of g at the fixed point: specifically, on $g'(L)$, where L is the root.

Procedure

- To sketch: draw $y = g(x)$ and $y = x$ (the line of gradient 1). Starting at x_0 on the x -axis, draw a vertical line up to $y=g(x)$ (giving $x_1 = g(x_0)$ as a y -value), then a horizontal line across to $y=x$ (transferring that value back onto the x -axis as the next x -value), and repeat.
- Convergence condition:** the iteration $x_{n+1}=g(x_n)$ converges to the root, from a starting point close enough to it, if $|g'(L)| < 1$ at the root L . If $|g'(L)| > 1$, the iteration diverges (moves away from the root).
- If $0 < g'(L) < 1$: staircase, converging monotonically from one side.
- If $-1 < g'(L) < 0$: cobweb, converging but oscillating (spiralling) around the root, alternating sides.

Worked example

EXAMPLE 9.3

For the iteration $x = (4x+2)^{1/3}$ from 9.2 (root at $L \approx 2.214$), find $g'(L)$ and hence state whether the iteration converges, and describe the shape of the diagram.

Step 1 $g(x) = (4x+2)^{1/3}$, so $g'(x) = \frac{4}{3} (4x+2)^{-2/3}$ (M1, using the chain rule from 7.7 with the general power rule).

Step 2 At $x = L \approx 2.214$: $4L+2 \approx 10.857$, so $g'(L) = 4/[3 \times 10.857^{2/3}] \approx 4/(3 \times 4.898) \approx 0.272$ (M1, A1).

Answer Since $0 < g'(L) \approx 0.272 < 1$, the iteration **converges**, and because $g'(L)$ is positive, the diagram is a **staircase** — the iterates approach the root steadily from the same side, without oscillating, which matches the values found in 9.2 (2, 2.154, 2.198, 2.210, ..., all increasing steadily toward 2.214) (A1).

COMMON MISTAKE

Only checking whether $g'(L)$ is small without checking its sign — the *magnitude* of $g'(L)$ determines whether it converges at all, but the *sign* determines the shape (staircase vs cobweb), and exam questions often ask for both.

9.4 The Newton–Raphson method

Concept

The Newton–Raphson method finds a root much faster than fixed-point iteration typically does, by using the tangent line to the curve as a local approximation. Starting from a guess x_0 , follow the tangent line at $(x_0, f(x_0))$ down to where it crosses the x -axis — that crossing point, x_1 , is usually a much better approximation to the root than x_0 was. Repeating this process converges extremely quickly (typically doubling the number of correct decimal places with each step) provided you start reasonably close to the root.

Procedure

The iteration formula (derived by setting the tangent line at x_n equal to zero and solving for the new x -value):

KEY FORMULA: NEWTON–RAPHSOIN ITERATION

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

In practice: find $f'(x)$ once, then repeatedly substitute the current approximation into this formula to generate the next one, continuing until successive values agree to the required accuracy.

Worked example

EXAMPLE 9.4

Use the Newton–Raphson method, with a starting value of $x_0 = 2.3$, to find the root of $x^3 - 4x - 2 = 0$ correct to 4 decimal places.

Step 1 $f(x) = x^3 - 4x - 2$, so $f'(x) = 3x^2 - 4$ (M1).

Step 2 Iteration formula: $x_{n+1} = x_n - (x_n^3 - 4x_n - 2)/(3x_n^2 - 4)$ (M1).

Step 3 $x_0 = 2.3$: $f(2.3) = 0.967$, $f'(2.3) = 11.87$. $x_1 = 2.3 - 0.967/11.87 = 2.2185$ (M1, A1 for one correctly worked iteration).

Step 4 $x_2 = 2.2185 - f(2.2185)/f'(2.2185) \approx 2.2143$. $x_3 \approx 2.21432$. $x_4 \approx 2.21432$ (agreeing to 5 d.p. already) (A1).

Answer So the root is **$x = 2.2143$ (4 d.p.)** — notice how much faster this converged than the fixed-point iteration in 9.2, which needed 5–6 steps for 3 decimal places; Newton–Raphson reached 4–5 decimal places in about 3 steps (A1).

n	0	1	2	3	4
x_n	2.3	2.2185	2.2143	2.21432	2.21432

COMMON MISTAKE

Making an arithmetic or sign error computing $f'(x_n)$ and carrying it through — since each iterate depends on the previous one, an error early on propagates and invalidates every subsequent value; it's worth double-checking $f'(x)$ itself (by differentiating carefully) before starting the iteration.

9.5 When numerical methods fail

Concept

Both fixed-point iteration and Newton–Raphson can fail to converge, or converge to the wrong root, or fail outright — and recognising when and why this happens is itself an examinable skill, not just a footnote. Understanding the failure modes deepens your understanding of why the methods work when they do.

Procedure — common failure modes to recognise

- **Newton–Raphson near a stationary point:** if $f'(x_n) = 0$ (or very close to zero) at some iterate, the formula involves dividing by (nearly) zero, sending the next iterate wildly off (or causing a calculator error) — this happens when the starting value is near a turning point of the curve.
- **Newton–Raphson with a poor starting value:** if x_0 is chosen far from the desired root, or near a different root/turning point, the method can converge to the *wrong* root, or fail to converge at all, oscillating or diverging instead.
- **Fixed-point iteration diverging:** as in 9.3, if $|g'(L)| > 1$ at the root for your chosen rearrangement $x = g(x)$, the iteration moves away from the root rather than toward it — the fix is

usually to choose a *different* algebraic rearrangement of the same equation, which will have a different g and a different $g'(L)$.

- **Discontinuities:** the change-of-sign method (9.1) can give a false positive across an asymptote, as noted in that section.

Worked example

EXAMPLE 9.5

Explain why applying the Newton–Raphson method to $f(x) = x^3 - 3x + 1$ with a starting value of $x_0 = 1$ fails.

Step 1 $f'(x) = 3x^2 - 3$ (M1).

Step 2 At $x_0 = 1$: $f'(1) = 3(1)^2 - 3 = 0$ (M1, A1).

Step 3 The Newton–Raphson formula requires dividing by $f'(x_0)$, but $f'(1) = 0$, so $x_1 = x_0 - f(x_0)/f'(x_0)$ is **undefined** (division by zero) (A1).

Answer Geometrically, $x=1$ is a stationary point of the curve, so the tangent line there is horizontal and never crosses the x -axis — there is no well-defined “next approximation” to follow (A1, geometric explanation).

COMMON MISTAKE

Assuming Newton–Raphson always works as long as you “just keep iterating” — always sanity-check that $f'(x_n)$ isn't zero or very small at each step, and that the sequence of iterates is actually settling down (not jumping around erratically) before trusting the final answer.

Chapter summary

Key ideas

- Change of sign: $f(a)$ and $f(b)$ opposite signs (f continuous) $\implies$ a root lies in (a,b) .
- Fixed-point iteration: rearrange $f(x)=0$ as $x=g(x)$, iterate $x_{n+1}=g(x_n)$.
- Convergence of fixed-point iteration at root L : needs $|g'(L)| < 1$; sign of $g'(L)$ determines staircase (positive) vs cobweb (negative).
- Newton–Raphson: $x_{n+1} = x_n - f(x_n)/f'(x_n)$; converges fast from a good starting value.
- Newton–Raphson fails when $f'(x_n) = 0$ or is close to zero, or when the starting value is poorly chosen.

Common exam mistakes across this chapter

- Concluding a root exists from a sign change without checking/stating continuity.
- Stopping an iteration before successive values have actually converged to the required accuracy.
- Checking only the magnitude of $g'(L)$, not its sign, when describing a cobweb/staircase diagram.
- Arithmetic slips in computing $f'(x_n)$ that then propagate through every later Newton–Raphson iterate.

-
- Not recognising or explaining why Newton–Raphson fails near a turning point.

You should now be able to

- Show a root lies in an interval using a change of sign, and narrow it down
- Carry out fixed-point iteration and judge convergence (and its shape) from $g'(L)$
- Apply the Newton–Raphson method, and recognise and explain when it fails